

DATA SHEET

TJA1050 High speed CAN transceiver

Preliminary specification
File under Integrated Circuits, IC18

1999 Sep 27

High speed CAN transceiver

TJA1050

FEATURES

- Fully compatible with the "ISO 11898" standard
- High speed (up to 1 Mbaud)
- Transmit Data (TXD) dominant time-out function
- Bus lines protected against transients in an automotive environment
- Silent mode in which the transmitter is disabled
- Differential receiver with wide common-mode range for high ElectroMagnetic Immunity (EMI)
- Input levels compatible with 3.3 V devices
- Thermally protected
- Short-circuit proof to battery and ground
- An unpowered node does not disturb the bus lines
- At least 110 nodes can be connected.

GENERAL DESCRIPTION

The TJA1050 is the interface between the CAN protocol controller and the physical bus. The device provides differential transmit capability to the bus and differential receive capability to the CAN controller.

The TJA1050 is the successor to the PCA82C250 high speed CAN transceiver. The most important improvements are:

- Much lower ElectroMagnetic Emission (EME) due to optimal matching of the CANH and CANL output signals
- Improved behaviour in case of an unpowered node.

QUICK REFERENCE DATA

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{CC}	supply voltage		4.75	5.25	V
V_{CANH}	DC voltage at CANH	$0 < V_{CC} < 5.25$ V; no time limit	-27	+40	V
V_{CANL}	DC voltage at CANL				
$V_{i(dif)(bus)}$	differential bus input voltage	dominant	1.5	3	V
$t_{PD(TXD-RXD)}$	propagation delay TXD to RXD; see Fig.4	$V_S = 0$ V	–	250	ns
T_{amb}	operating ambient temperature		-40	+125	°C

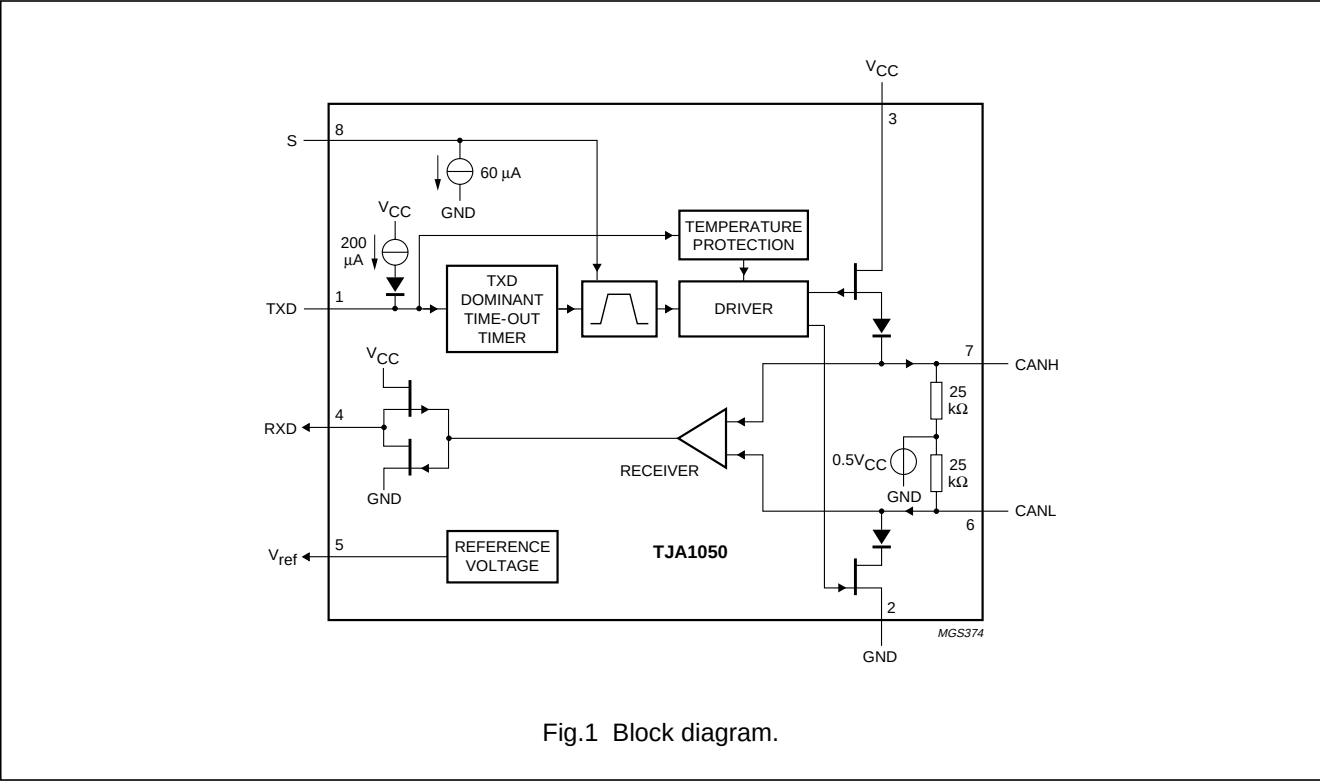
ORDERING INFORMATION

TYPE NUMBER	PACKAGE		
	NAME	DESCRIPTION	VERSION
TJA1050T	SO8	plastic small outline package; 8 leads; body width 3.9 mm	SOT96-1
TJA1050U	–	bare die	–

High speed CAN transceiver

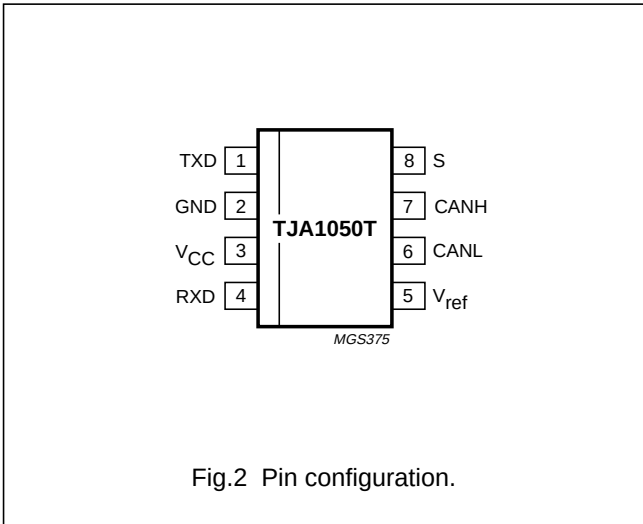
TJA1050

BLOCK DIAGRAM



PINNING

SYMBOL	PIN	DESCRIPTION
TXD	1	transmit data input; reads in data from the CAN controller to the bus line drivers
GND	2	ground
V _{CC}	3	supply voltage
RXD	4	receive data output; reads out data from the bus lines to the CAN controller
V _{ref}	5	reference voltage output
CANL	6	LOW-level CAN bus line
CANH	7	HIGH-level CAN bus line
S	8	select input for high speed mode/silent mode



High speed CAN transceiver

TJA1050

FUNCTIONAL DESCRIPTION

The TJA1050 is the interface between the CAN protocol controller and the physical bus. It is primarily intended for high speed automotive applications using baud rates from 40 kbaud up to 1 Mbaud. It provides differential transmit capability to the bus and differential receiver capability to the CAN protocol controller. It is fully compatible to the "ISO 11898" standard.

A current-limiting circuit protects the transmitter output stage from damage caused by accidental short-circuit to either positive or negative battery voltage, although power dissipation increases during this fault condition.

A thermal protection circuit protects the IC from damage by switching off the transmitter if the junction temperature exceeds a value of approximately 165 °C. Because the transmitter dissipates most of the power, the power dissipation and temperature of the IC is reduced. All other IC functions continue to operate. The transmitter off-state resets when TXD goes HIGH. The thermal protection circuit is particularly needed when a bus line short-circuits.

The CANH and CANL lines are protected from automotive electrical transients (according to "ISO 7637", see Fig.6) and are also protected from Electro-Static-Discharge (ESD) of up to 4 kV from the human body.

Control line S (pin 8) allows two operating modes to be selected; high speed mode or silent mode.

High speed mode is the normal operating mode and is selected by connecting pin S to ground. It is the default mode if pin S is unconnected.

In the silent mode, the transmitter is disabled. All other IC functions continue to operate. The silent mode is selected by connecting pin S to V_{CC} .

A 'TXD Dominant Time-out' timer circuit prevents the bus lines being driven to a permanent dominant state (blocking all network communication) if TXD is forced permanently LOW by a hardware and/or software application failure. The timer is triggered by a negative edge on TXD. If the duration of the LOW-level on TXD exceeds the internal timer value, the transmitter is disabled, driving the bus into a recessive state. The timer is reset by a positive edge on TXD.

Table 1 Function table of the CAN transceiver
(X = don't care)

V_{CC}	TXD	S	CANH	CANL	BUS STATE	RXD
4.75 to 5.25 V	0	0 (or floating)	HIGH	LOW	dominant	0
4.75 to 5.25 V	X	1	$0.5 \times V_{CC}$	$0.5 \times V_{CC}$	recessive	1
4.75 to 5.25 V	1 (or floating)	X	$0.5 \times V_{CC}$	$0.5 \times V_{CC}$	recessive	1
<2 V (not powered)	X	X	0 V < CANH < V_{CC}	0 V < CANL < V_{CC}	recessive	X
2 V < V_{CC} < 4.75 V	>2 V	X	0 V < CANH < V_{CC}	0 V < CANL < V_{CC}	recessive	X

High speed CAN transceiver

TJA1050

LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134). All voltages are referenced to GND (pin 2). Positive currents flow into the IC.

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{CC}	supply voltage		-0.3	+5.25	V
V_{CANL} , V_{CANH}	DC voltage at CANL and CANH	$0 < V_{CC} < 5.25$ V; no time limit	-27	+40	V
V_{TXD} , V_{RXD} , V_{ref} and V_S	DC voltage at TXD, RXD, V_{ref} and S		-0.3	$V_{CC} + 0.3$	V
$V_{trt(CANH)}$, $V_{trt(CANL)}$	transient voltage at CANH and CANL	time limit is 1 μ s	-55	+55	V
		note 1	-200	+200	V
V_{esd}	electrostatic discharge at CANH; CANL	note 3	-4	+4	kV
	electrostatic discharge at TXD; V_{CC} ; RXD; V_{ref} and S	note 3	-2	+2	kV
	electrostatic discharge at all pins	note 4	-200	+200	V
T_{stg}	storage temperature		-55	+150	$^{\circ}$ C
T_{amb}	operating ambient temperature		-40	+125	$^{\circ}$ C
T_j	junction temperature	note 2	-40	+150	$^{\circ}$ C

Notes

1. The waveforms of the applied transients shall be in accordance with "ISO 7637 part 1", test pulses 1, 2, 3a and 3b, (see Fig.6).
2. In accordance with "IEC 747-1". An alternative definition of T_j is: $T_j = T_{amb} + P \times R_{th(j-a)}$, where $R_{th(j-a)}$ is a fixed value to be used for the calculation of T_j . The rating for T_j limits the allowable combinations of power dissipation (P) and ambient temperature (T_{amb}).
3. Human body model; C = 100 pF R = 1.5 k Ω .
4. Machine model; C = 200 pF R = 25 Ω .

THERMAL CHARACTERISTICS

According to IEC 747-1.

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
$R_{th(j-a)}$	thermal resistance from junction to ambient; TJA1050T(SO8)	in free air	160	K/W

QUALITY SPECIFICATION

Quality specification "SNW-FQ-611 part D" is applicable.

High speed CAN transceiver

TJA1050

CHARACTERISTICS

$V_{CC} = 4.75$ to 5.25 V; $T_{amb} = -40$ to $+125$ °C; $R_L = 60$ Ω unless specified otherwise; all voltages are referenced to GND (pin 2); positive currents flow into the IC; all parameters are guaranteed over the ambient temperature range by design, but only 100% tested at $T_{amb} = 25$ °C unless specified otherwise.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Supply (V _{CC})						
I _{CC}	supply current	dominant; V _{TXD} = 0 V	tbf	–	75	mA
		recessive; V _{TXD} = V _{CC}	tbf	–	13	mA
Transmitter data input (TXD)						
V _{IH}	HIGH-level input voltage	output recessive	2.0	–	V _{CC} + 0.3	V
V _{IL}	LOW-level input voltage	output dominant	–0.3	–	+0.8	V
I _{IH}	HIGH-level input current	V _{TXD} = V _{CC}	–30	0	+30	μA
I _{IL}	LOW-level input current	V _{TXD} = 0 V	–100	–200	–300	μA
C _{i(TXD)}	TXD input capacitance	not tested	–	–	tbf	pF
Mode select input (S)						
V _{IH}	HIGH-level input voltage	silent mode	2.0	–	V _{CC} + 0.3	V
V _{IL}	LOW-level input voltage	high speed mode	–0.3	–	+0.8	V
I _{IH}	HIGH-level input current	V _S = V _{CC}	30	60	100	μA
I _{IL}	LOW-level input current	V _S = 0 V	–30	0	+30	μA
Receiver data output (RXD)						
I _{OH}	HIGH-level output current	V _{RXD} = 0.7 V _{CC}	tbf	tbf	tbf	mA
I _{OL}	LOW-level output current	V _{RXD} = 0.45 V	2	8.5	20	mA
V _{ref}						
V _{ref}	reference output voltage	–50 μA < I _{Vref} < 50 μA	0.45V _{CC}	0.5V _{CC}	0.55V _{CC}	V
Bus lines (CANH; CANL)						
V _{CANH(reces)} ; V _{CANL(reces)}	recessive bus voltage	V _{TXD} = V _{CC} ; no load	2.0	–	3.0	V
I _{o(CANH)(reces)} ; I _{o(CANL)(reces)}	recessive output current	–27 V < V _{CANH} , V _{CANL} < 32 V; 0 V < V _{CC} < 5.25 V	–2.5	–	+2.5	mA
V _{o(CANH)}	CANH dominant output voltage	V _{TXD} = 0 V	2.8	–	4.5	V
V _{o(CANL)}	CANL dominant output voltage		0.5	–	2.0	V
V _{i(dif)(bus)}	differential bus input voltage (V _{CANH} – V _{CANL})	V _{TXD} = 0 V; 42.5 < R _L < 60 Ω (dominant)	1.5	–	3.0	V
		V _{TXD} = V _{CC} ; no load (recessive)	–500	–	+50	mV
I _{o(sc)(CANH)}	CANH short-circuit output current	V _{CANH} = 0 V; V _{TXD} = 0 V	–35	–	–95	mA
I _{o(sc)(CANL)}	CANL short-circuit output current	V _{CANL} = 36 V; V _{TXD} = 0 V	35	–	150	mA

High speed CAN transceiver

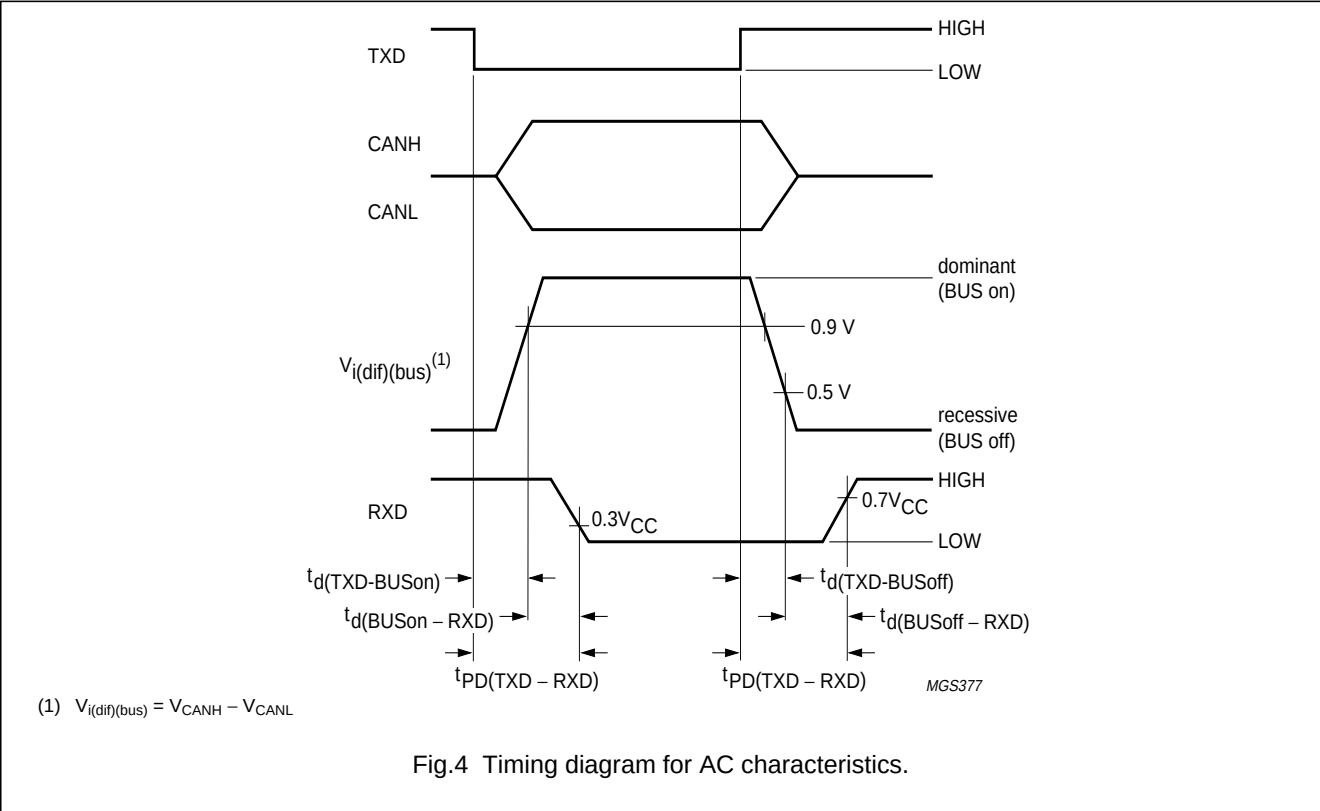
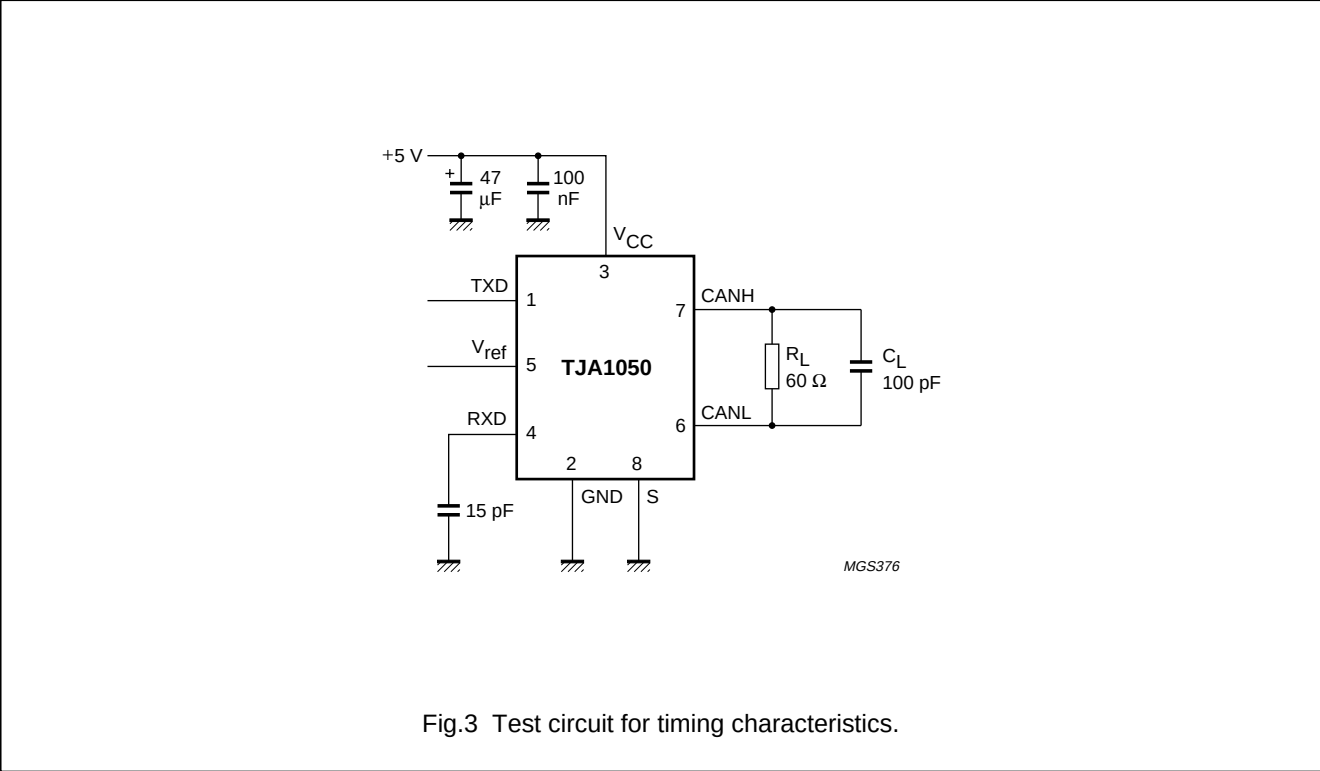
TJA1050

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
$V_{dif(th)}$	differential receiver threshold voltage	$-12\text{ V} < V_{CANH}$, $V_{CANL} < 12\text{ V}$; see Fig.5	0.5	0.7	0.9	V
$V_{i(dif)(hys)}$	differential receiver input voltage hysteresis	see Fig.5	100	–	200	mV
$R_{i(cm)(CANH)}$; $R_{i(cm)(CANL)}$	CANH; CANL common mode input resistance		10	25	50	k Ω
$R_{i(cm)(m)}$	matching between CANH and CANL common mode input resistance	$V_{CANH} = V_{CANL}$	–3	–	+3	%
$R_{i(dif)}$	differential input resistance		20	50	100	k Ω
$C_{i(CANH)}$; $C_{i(CANL)}$	CANH; CANL input capacitance	$V_{TXD} = V_{CC}$; not tested	–	–	20	pF
$C_{i(dif)}$	differential input capacitance		–	–	10	pF
$I_{LI(CANH)}$; $I_{LI(CANL)}$	CANH; CANL input leakage current	$V_{CC} = 0\text{ V}$; $V_{CANH} = V_{CANL} = 5\text{ V}$	–	–	500	μA
Thermal shutdown						
$T_{j(sd)}$	shutdown junction temperature		155	165	180	$^{\circ}\text{C}$
Timing characteristics (see Figs 3 and 4)						
$t_{d(TXD-BUSon)}$	delay TXD to bus active	$V_S = 0\text{ V}$	tbf	tbf	150	ns
$t_{d(TXD-BUSoff)}$	delay TXD to bus inactive					
$t_{d(BUSon-RXD)}$	delay bus active to RXD		tbf	tbf	100	ns
$t_{d(BUSoff-RXD)}$	delay bus inactive to RXD					

High speed CAN transceiver

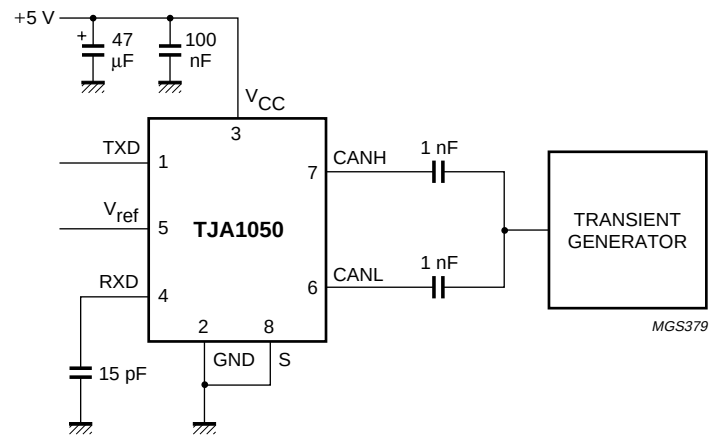
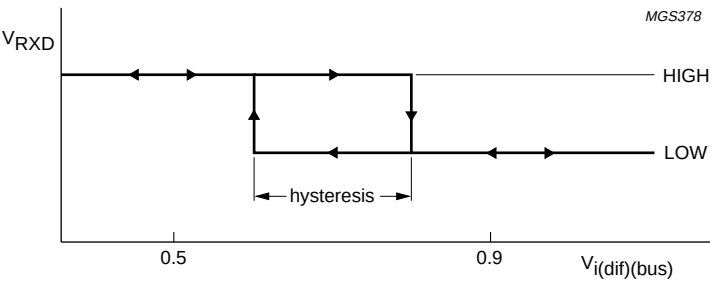
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TEST AND APPLICATION INFORMATION



High speed CAN transceiver

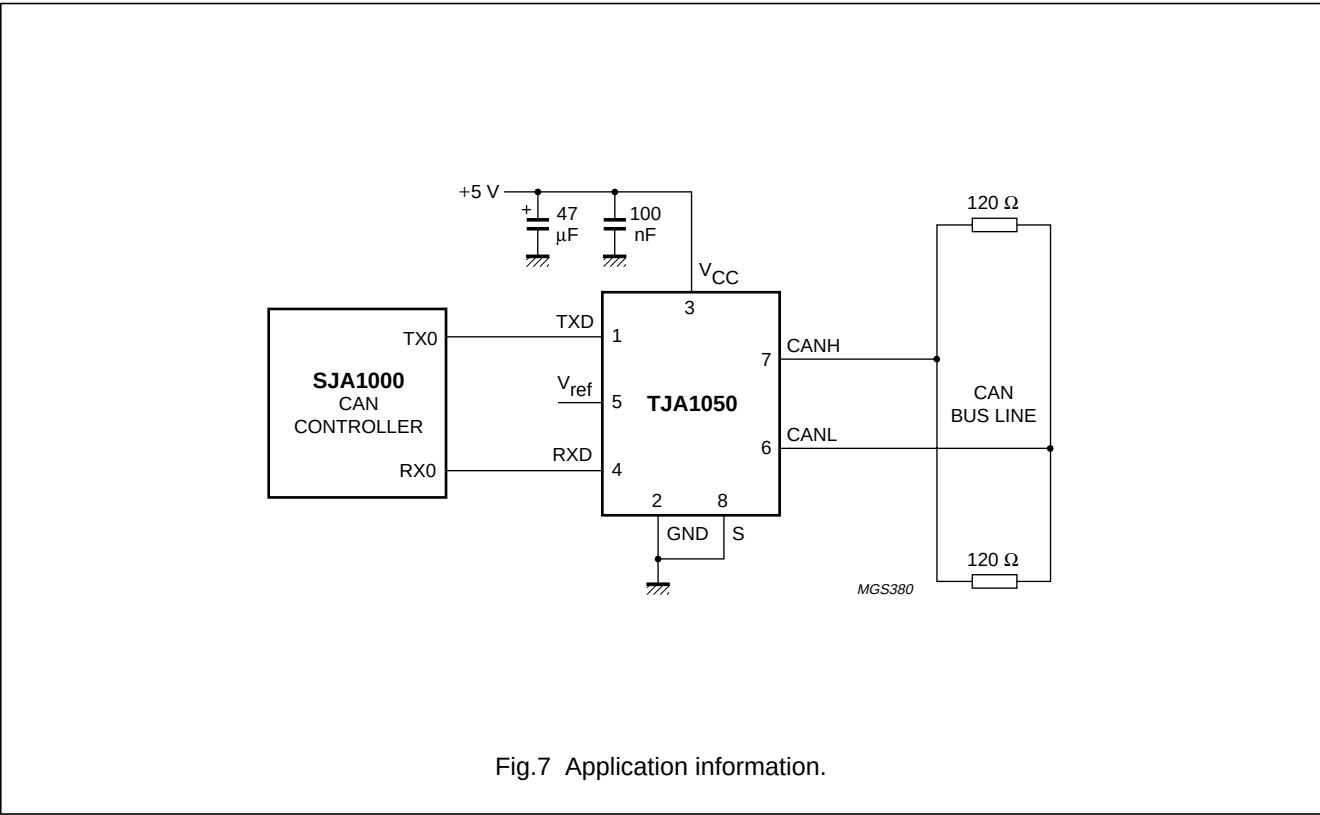
TJA1050



The waveforms of the applied transients shall be in accordance with "ISO 7637 part 1", test pulses 1, 2, 3a and 3b.

High speed CAN transceiver

TJA1050



BONDING PAD LOCATIONS FOR TJA1050U

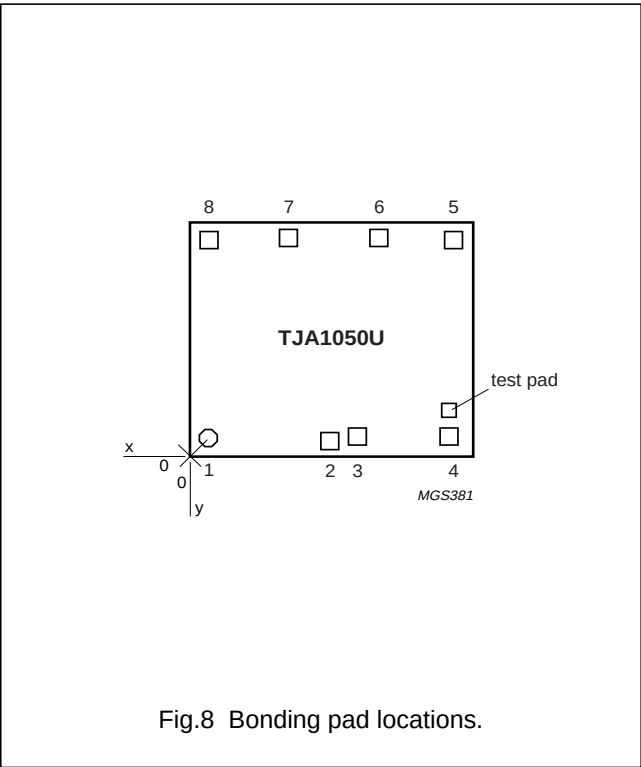


Table 2 Bonding pad locations

All x/y coordinates represent the position of the centre of each pad (in μm) with respect to $x/y = 0$ of the die (see Fig.8).

SYMBOL	PAD	COORDINATES	
		x	y
TXD	1	103	103
GND	2	740.5	85
V _{CC}	3	886.5	111
RXD	4	1371.5	111
V _{ref}	5	1394	1094
CANL	6	1006	1111
CANH	7	542.5	1111
S	8	103	1097

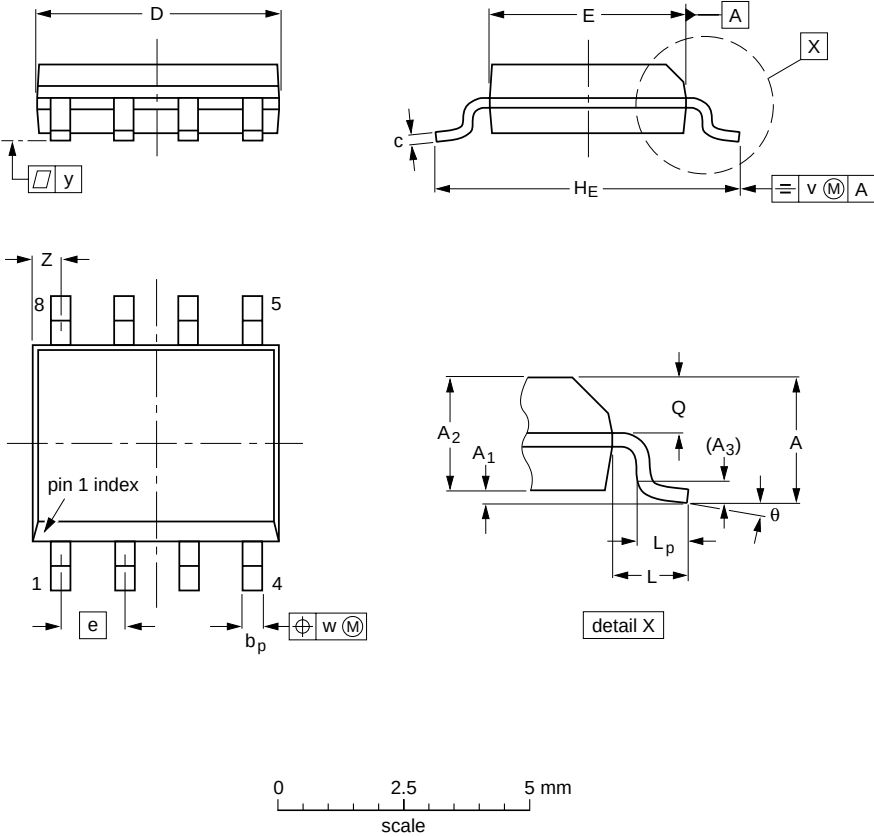
High speed CAN transceiver

TJA1050

PACKAGE OUTLINE

SO8: plastic small outline package; 8 leads; body width 3.9 mm

SOT96-1



DIMENSIONS (inch dimensions are derived from the original mm dimensions)

UNIT	A max.	A ₁	A ₂	A ₃	b _p	c	D ⁽¹⁾	E ⁽²⁾	e	H _E	L	L _p	Q	v	w	y	z ⁽¹⁾	θ
mm	1.75	0.25 0.10	1.45 1.25	0.25	0.49 0.36	0.25 0.19	5.0 4.8	4.0 3.8	1.27	6.2 5.8	1.05	1.0 0.4	0.7 0.6	0.25	0.25	0.1	0.7 0.3	8° 0°
inches	0.069	0.010 0.004	0.057 0.049	0.01	0.019 0.014	0.0100 0.0075	0.20 0.19	0.16 0.15	0.050	0.244 0.228	0.041	0.039 0.016	0.028 0.024	0.01	0.01	0.004	0.028 0.012	

Notes

- 1. Plastic or metal protrusions of 0.15 mm maximum per side are not included.
- 2. Plastic or metal protrusions of 0.25 mm maximum per side are not included.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT96-1	076E03S	MS-012AA				95-02-04 97-05-22

High speed CAN transceiver

TJA1050

SOLDERING

Introduction to soldering surface mount packages

This text gives a very brief insight to a complex technology. A more in-depth account of soldering ICs can be found in our *"Data Handbook IC26; Integrated Circuit Packages"* (document order number 9398 652 90011).

There is no soldering method that is ideal for all surface mount IC packages. Wave soldering is not always suitable for surface mount ICs, or for printed-circuit boards with high population densities. In these situations reflow soldering is often used.

Reflow soldering

Reflow soldering requires solder paste (a suspension of fine solder particles, flux and binding agent) to be applied to the printed-circuit board by screen printing, stencilling or pressure-syringe dispensing before package placement.

Several methods exist for reflowing; for example, infrared/convection heating in a conveyor type oven. Throughput times (preheating, soldering and cooling) vary between 100 and 200 seconds depending on heating method.

Typical reflow peak temperatures range from 215 to 250 °C. The top-surface temperature of the packages should preferably be kept below 230 °C.

Wave soldering

Conventional single wave soldering is not recommended for surface mount devices (SMDs) or printed-circuit boards with a high component density, as solder bridging and non-wetting can present major problems.

To overcome these problems the double-wave soldering method was specifically developed.

If wave soldering is used the following conditions must be observed for optimal results:

- Use a double-wave soldering method comprising a turbulent wave with high upward pressure followed by a smooth laminar wave.
- For packages with leads on two sides and a pitch (e):
 - larger than or equal to 1.27 mm, the footprint longitudinal axis is **preferred** to be parallel to the transport direction of the printed-circuit board;
 - smaller than 1.27 mm, the footprint longitudinal axis **must** be parallel to the transport direction of the printed-circuit board.

The footprint must incorporate solder thieves at the downstream end.

- For packages with leads on four sides, the footprint must be placed at a 45° angle to the transport direction of the printed-circuit board. The footprint must incorporate solder thieves downstream and at the side corners.

During placement and before soldering, the package must be fixed with a droplet of adhesive. The adhesive can be applied by screen printing, pin transfer or syringe dispensing. The package can be soldered after the adhesive is cured.

Typical dwell time is 4 seconds at 250 °C.

A mildly-activated flux will eliminate the need for removal of corrosive residues in most applications.

Manual soldering

Fix the component by first soldering two diagonally-opposite end leads. Use a low voltage (24 V or less) soldering iron applied to the flat part of the lead. Contact time must be limited to 10 seconds at up to 300 °C.

When using a dedicated tool, all other leads can be soldered in one operation within 2 to 5 seconds between 270 and 320 °C.

High speed CAN transceiver

TJA1050

Suitability of surface mount IC packages for wave and reflow soldering methods

PACKAGE	SOLDERING METHOD	
	WAVE	REFLOW ⁽¹⁾
BGA, SQFP	not suitable	suitable
HLQFP, HSQFP, HSOP, HTSSOP, SMS	not suitable ⁽²⁾	suitable
PLCC ⁽³⁾ , SO, SOJ	suitable	suitable
LQFP, QFP, TQFP	not recommended ⁽³⁾⁽⁴⁾	suitable
SSOP, TSSOP, VSO	not recommended ⁽⁵⁾	suitable

Notes

1. All surface mount (SMD) packages are moisture sensitive. Depending upon the moisture content, the maximum temperature (with respect to time) and body size of the package, there is a risk that internal or external package cracks may occur due to vaporization of the moisture in them (the so called popcorn effect). For details, refer to the Drypack information in the *"Data Handbook IC26; Integrated Circuit Packages; Section: Packing Methods"*.
2. These packages are not suitable for wave soldering as a solder joint between the printed-circuit board and heatsink (at bottom version) can not be achieved, and as solder may stick to the heatsink (on top version).
3. If wave soldering is considered, then the package must be placed at a 45° angle to the solder wave direction. The package footprint must incorporate solder thieves downstream and at the side corners.
4. Wave soldering is only suitable for LQFP, TQFP and QFP packages with a pitch (e) equal to or larger than 0.8 mm; it is definitely not suitable for packages with a pitch (e) equal to or smaller than 0.65 mm.
5. Wave soldering is only suitable for SSOP and TSSOP packages with a pitch (e) equal to or larger than 0.65 mm; it is definitely not suitable for packages with a pitch (e) equal to or smaller than 0.5 mm.

High speed CAN transceiver

TJA1050

DEFINITIONS

Data sheet status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

LIFE SUPPORT APPLICATIONS

These products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury. Philips customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Philips for any damages resulting from such improper use or sale.

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TJA1050

NOTES

Philips Semiconductors – a worldwide company

Argentina: see South America

Australia: 3 Figtree Drive, HOMEBUSH, NSW 2140,
Tel. +61 2 9704 8141, Fax. +61 2 9704 8139

Austria: Computerstr. 6, A-1101 WIEN, P.O. Box 213,
Tel. +43 1 60 101 1248, Fax. +43 1 60 101 1210

Belarus: Hotel Minsk Business Center, Bld. 3, r. 1211, Volodarski Str. 6,
220050 MINSK, Tel. +375 172 20 0733, Fax. +375 172 20 0773

Belgium: see The Netherlands

Brazil: see South America

Bulgaria: Philips Bulgaria Ltd., Energoproject, 15th floor,
51 James Bourchier Blvd., 1407 SOFIA,
Tel. +359 2 68 9211, Fax. +359 2 68 9102

Canada: PHILIPS SEMICONDUCTORS/COMPONENTS,
Tel. +1 800 234 7381, Fax. +1 800 943 0087

China/Hong Kong: 501 Hong Kong Industrial Technology Centre,
72 Tat Chee Avenue, Kowloon Tong, HONG KONG,
Tel. +852 2319 7888, Fax. +852 2319 7700

Colombia: see South America

Czech Republic: see Austria

Denmark: Sydhavnsgade 23, 1780 COPENHAGEN V,
Tel. +45 33 29 3333, Fax. +45 33 29 3905

Finland: Sinikalliontie 3, FIN-02630 ESPOO,
Tel. +358 9 615 800, Fax. +358 9 6158 0920

France: 51 Rue Carnot, BP317, 92156 SURESNES Cedex,
Tel. +33 1 4099 6161, Fax. +33 1 4099 6427

Germany: Hammerbrookstraße 69, D-20097 HAMBURG,
Tel. +49 40 2353 60, Fax. +49 40 2353 6300

Hungary: see Austria

India: Philips INDIA Ltd, Band Box Building, 2nd floor,
254-D, Dr. Annie Besant Road, Worli, MUMBAI 400 025,
Tel. +91 22 493 8541, Fax. +91 22 493 0966

Indonesia: PT Philips Development Corporation, Semiconductors Division,
Gedung Philips, Jl. Buncit Raya Kav.99-100, JAKARTA 12510,
Tel. +62 21 794 0040 ext. 2501, Fax. +62 21 794 0080

Ireland: Newstead, Clonskeagh, DUBLIN 14,
Tel. +353 1 7640 000, Fax. +353 1 7640 200

Israel: RAPAC Electronics, 7 Kehilat Saloniki St, PO Box 18053,
TEL AVIV 61180, Tel. +972 3 645 0444, Fax. +972 3 649 1007

Italy: PHILIPS SEMICONDUCTORS, Via Casati, 23 - 20052 MONZA (MI),
Tel. +39 039 203 6838, Fax +39 039 203 6800

Japan: Philips Bldg 13-37, Kohnan 2-chome, Minato-ku,
TOKYO 108-8507, Tel. +81 3 3740 5130, Fax. +81 3 3740 5057

Korea: Philips House, 260-199 Itaewon-dong, Yongsan-ku, SEOUL,
Tel. +82 2 709 1412, Fax. +82 2 709 1415

Malaysia: No. 76 Jalan Universiti, 46200 PETALING JAYA, SELANGOR,
Tel. +60 3 750 5214, Fax. +60 3 757 4880

Mexico: 5900 Gateway East, Suite 200, EL PASO, TEXAS 79905,
Tel. +9-5 800 234 7381, Fax +9-5 800 943 0087

Middle East: see Italy

Netherlands: Postbus 90050, 5600 PB EINDHOVEN, Bldg. VB,
Tel. +31 40 27 82785, Fax. +31 40 27 88399

New Zealand: 2 Wagener Place, C.P.O. Box 1041, AUCKLAND,
Tel. +64 9 849 4160, Fax. +64 9 849 7811

Norway: Box 1, Manglerud 0612, OSLO,
Tel. +47 22 74 8000, Fax. +47 22 74 8341

Pakistan: see Singapore

Philippines: Philips Semiconductors Philippines Inc.,
106 Valero St. Salcedo Village, P.O. Box 2108 MCC, MAKATI,
Metro MANILA, Tel. +63 2 816 6380, Fax. +63 2 817 3474

Poland: Al.Jerozolimskie 195 B, 02-222 WARSAW,
Tel. +48 22 5710 000, Fax. +48 22 5710 001

Portugal: see Spain

Romania: see Italy

Russia: Philips Russia, Ul. Usatcheva 35A, 119048 MOSCOW,
Tel. +7 095 755 6918, Fax. +7 095 755 6919

Singapore: Lorong 1, Toa Payoh, SINGAPORE 319762,
Tel. +65 350 2538, Fax. +65 251 6500

Slovakia: see Austria

Slovenia: see Italy

South Africa: S.A. PHILIPS Pty Ltd., 195-215 Main Road Martindale,
2092 JOHANNESBURG, P.O. Box 58088 Newville 2114,
Tel. +27 11 471 5401, Fax. +27 11 471 5398

South America: Al. Vicente Pinzon, 173, 6th floor,
04547-130 SÃO PAULO, SP, Brazil,
Tel. +55 11 821 2333, Fax. +55 11 821 2382

Spain: Balmes 22, 08007 BARCELONA,
Tel. +34 93 301 6312, Fax. +34 93 301 4107

Sweden: Kottbygatan 7, Akalla, S-16485 STOCKHOLM,
Tel. +46 8 5985 2000, Fax. +46 8 5985 2745

Switzerland: Allmendstrasse 140, CH-8027 ZÜRICH,
Tel. +41 1 488 2741 Fax. +41 1 488 3263

Taiwan: Philips Semiconductors, 6F, No. 96, Chien Kuo N. Rd., Sec. 1,
TAIPEI, Taiwan Tel. +886 2 2134 2886, Fax. +886 2 2134 2874

Thailand: PHILIPS ELECTRONICS (THAILAND) Ltd.,
209/2 Sanpavuth-Bangna Road Prakanong, BANGKOK 10260,
Tel. +66 2 745 4090, Fax. +66 2 398 0793

Turkey: Yukari Dudullu, Org. San. Blg., 2.Cad. Nr. 28 81260 Umraniye,
ISTANBUL, Tel. +90 216 522 1500, Fax. +90 216 522 1813

Ukraine: PHILIPS UKRAINE, 4 Patrice Lumumba str., Building B, Floor 7,
252042 KIEV, Tel. +380 44 264 2776, Fax. +380 44 268 0461

United Kingdom: Philips Semiconductors Ltd., 276 Bath Road, Hayes,
MIDDLESEX UB3 5BX, Tel. +44 208 730 5000, Fax. +44 208 754 8421

United States: 811 East Arques Avenue, SUNNYVALE, CA 94088-3409,
Tel. +1 800 234 7381, Fax. +1 800 943 0087

Uruguay: see South America

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Yugoslavia: PHILIPS, Trg N. Pasica 5/v, 11000 BEOGRAD,
Tel. +381 11 62 5344, Fax.+381 11 63 5777

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SCA 68

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